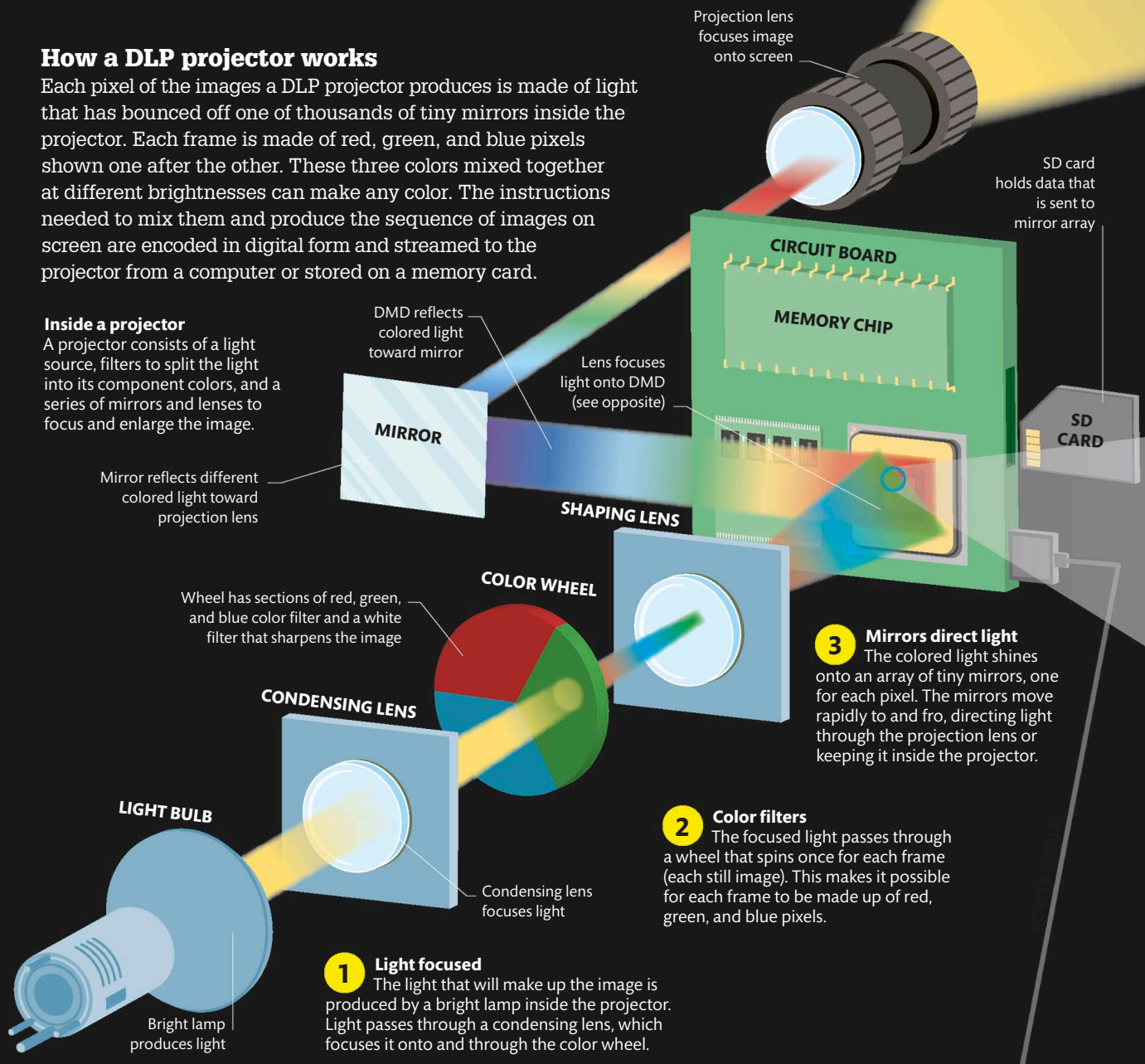
Projectors

A projector produces 25, 30, or 60 bright images on a screen each second. Each image, or frame, is composed of thousands of pixels. There are a few different ways of producing the pixels, but the most common projector technology is DLP: digital light processing.

How a DLP projector works

Each pixel of the images a DLP projector produces is made of light that has bounced off one of thousands of tiny mirrors inside the projector. Each frame is made of red, green, and blue pixels shown one after the other. These three colors mixed together at different brightnesses can make any color. The instructions needed to mix them and produce the sequence of images on screen are encoded in digital form and streamed to the projector from a computer or stored on a memory card.

Inside a projector
A projector consists of a light source, filters to split the light into its component colors, and a series of mirrors and lenses to focus and enlarge the image.



4 Image projected
Any light the mirrors have diverted through the lens is focused onto a screen. The light from all the mirrors makes up the projected image.

3 Mirrors direct light
The colored light shines onto an array of tiny mirrors, one for each pixel. The mirrors move rapidly to and fro, directing light through the projection lens or keeping it inside the projector.

2 Color filters
The focused light passes through a wheel that spins once for each frame (each still image). This makes it possible for each frame to be made up of red, green, and blue pixels.

1 Light focused
The light that will make up the image is produced by a bright lamp inside the projector. Light passes through a condensing lens, which focuses it onto and through the color wheel.